

Computer Graphics and Visualization

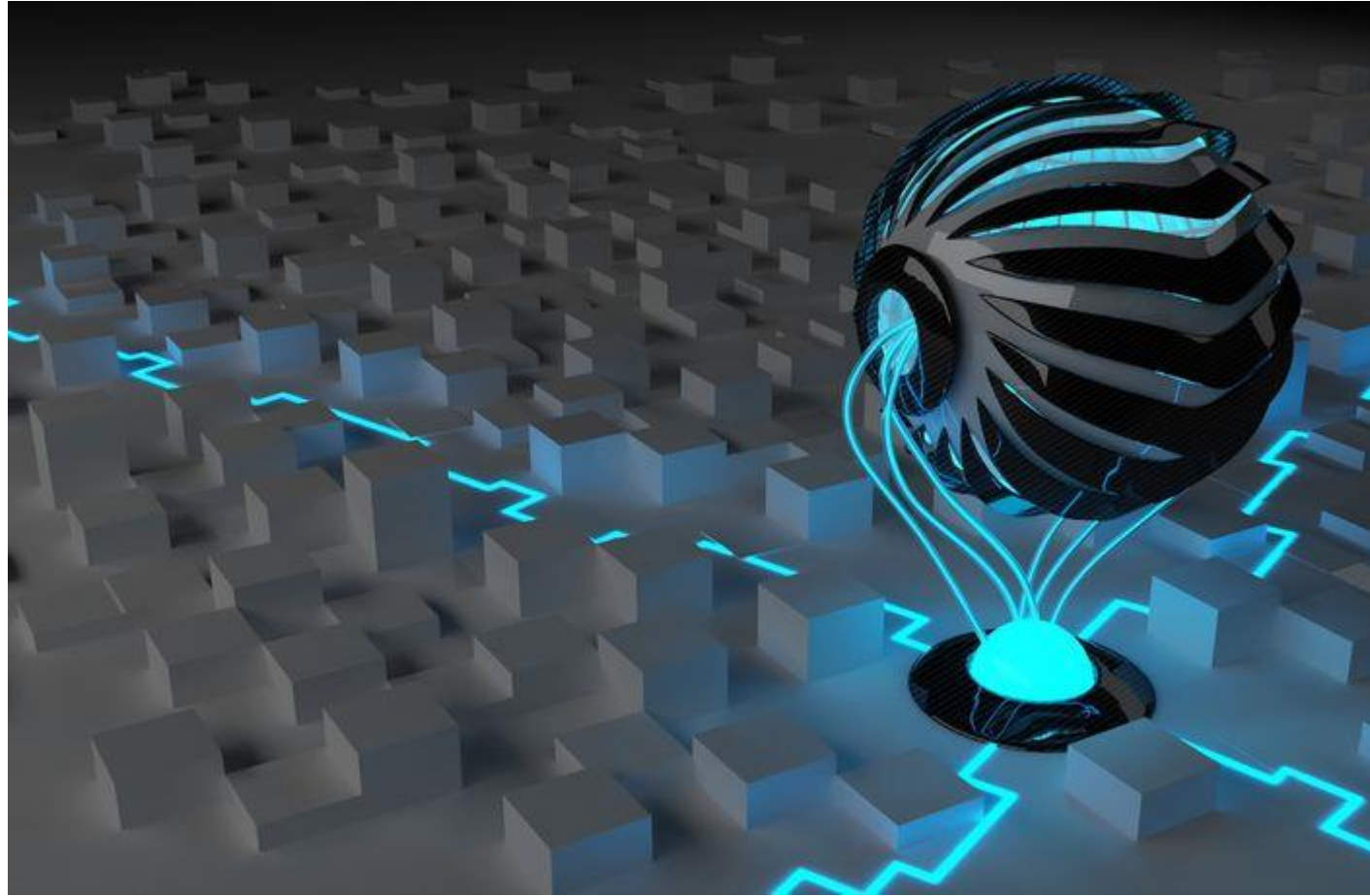
Chapter 7

Computer Animation and Visualization

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Computer Animation and Visualization

7.1. Computer Animation functions

7.2. Raster Animations

7.3. Key-Frame Systems

7.4. Motion Specification

7.4.1. Direct-Motion Specification

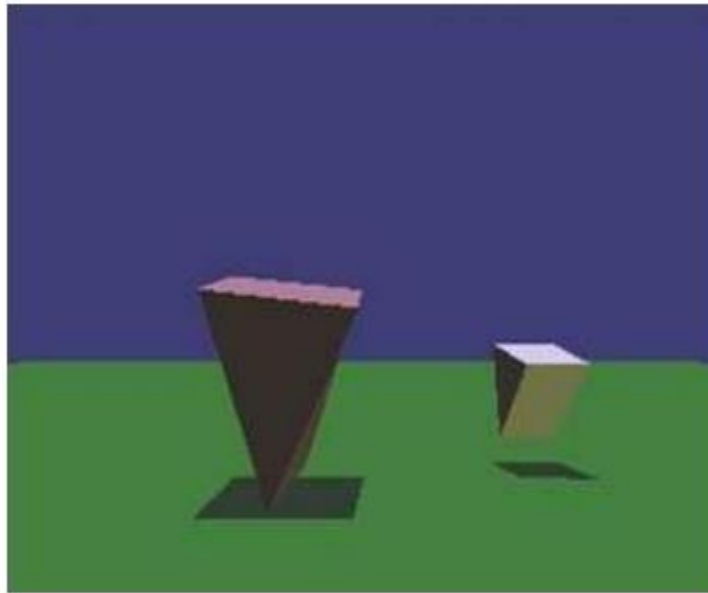
7.4.2. Goal-Directed Systems

7.4.3. Kinematics and Dynamics

Computer Animation

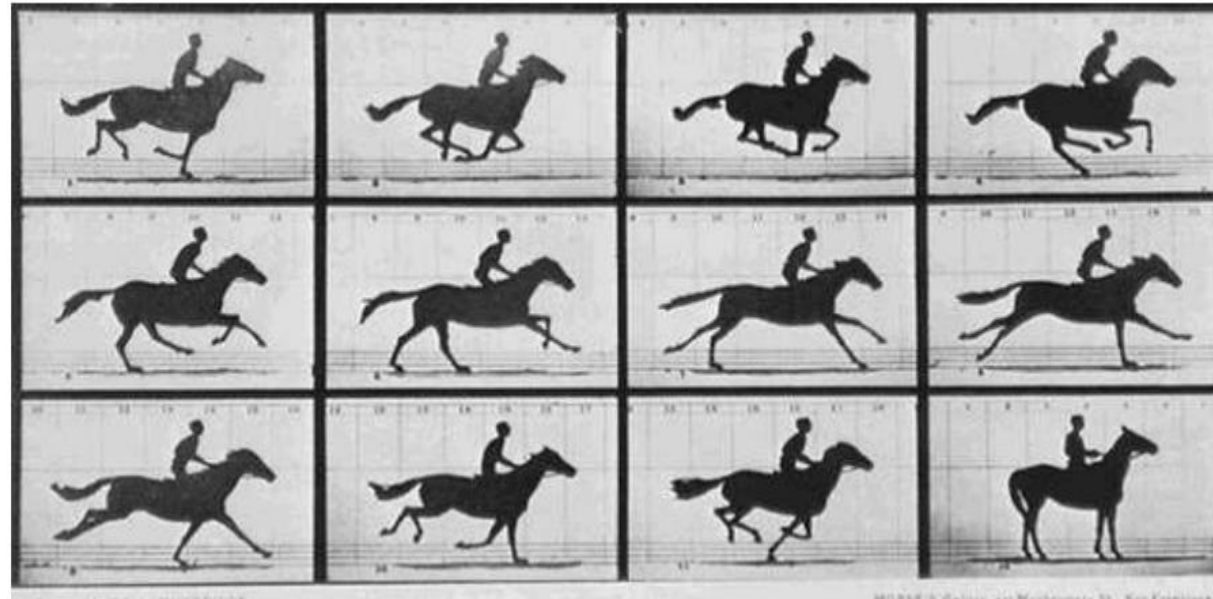
Overview

Motion can bring the simplest of characters to life. Even simple polygonal shapes can convey a number of human qualities when animated: identity, character, gender, mood, intention, emotion, and so on. Very simple



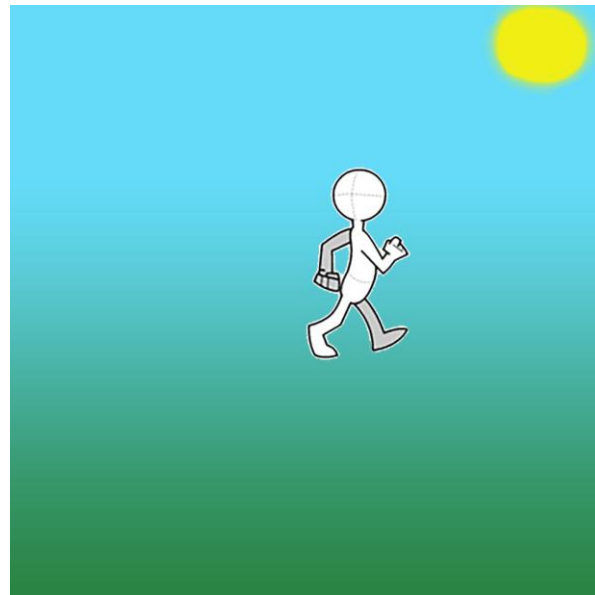
Very simple characters

- A movie is a sequence of frames of still images. For video, the frame rate is typically 24 frames per second. For film, this is 30 frames per second.



The horse in motion

- ❑ Some typical applications of **computer-generated animation** are entertainment (motion pictures and cartoons), advertising, scientific and engineering studies, and training and education. Although we tend to think of animation as implying object motions, the term computer animation generally refers to any time sequence of visual changes in a scene.
- ❑ In addition to changing object position with translations or rotations, a computer-generated animation could display time variations in object size, color, transparency, or surface texture.
- ❑ Advertising animations often transition one object shape into another: for example, transforming a can of motor oil into an automobile engine.
- ❑ Computer animations can also be generated by changing camera parameters, such as position, orientation, and focal length. And we can produce computer animations by changing lighting effects or other parameters and procedures associated with illumination and rendering



DESIGN OF ANIMATION SEQUENCES

In general, an animation sequence is designed with the following steps:

- ❑ Storyboard layout
- ❑ Object definitions
- ❑ Key-frame specification
- ❑ Generation of in-between frames

- ❑ The **storyboard** is an outline of the action. It defines the motion sequence as a set of basic events that are to take place. Depending on the type of animation to be produced, the storyboard could consist of a set of rough sketches or it could be a list of the basic ideas for the motion.
- ❑ An **object definition** is given for each participant in the action. Objects can be defined in terms of basic shapes, such as polygons or splines. In addition, the associated movements for each object are specified along with the shape
- ❑ A **key-frame** is a detailed drawing of the scene at a certain time in the animation sequence. Within each key frame, each object is positioned according to the time for that frame. Some key frames are chosen at extreme positions in the action; others are spaced so that the time interval between key frames is not too great. More key frames are specified for intricate motions than for simple, slowly varying motions.
- ❑ **In-between** are the intermediate frames between **the key frames**. The number of in-betweens needed is determined by the media to be used to display the animation. Film requires 24 frames per second, and graphics terminals are refreshed at the rate of 30 to 60 frames per second. Typically, time intervals for the motion are set up so that there are from three to five in-betweens for each pair of key frames. Depending on the speed specified for the motion, some key frames can be duplicated.

- Eg. For a 1 minute film sequence with no duplication, we would need 1440 frames. With five in-betweens for each pair of key frames, we would need 288 key frames. If the motion is not too complicated, we could space the key frames a little farther apart.

There are several other tasks that may be required, depending on the application. They include motion verification, editing, and production and synchronization of a soundtrack. Many of the functions needed to produce general animations are now computer-generated

Example

Story Layout Board

- a. In our animation one apple falls from a tree and bounces to the ground.
- b. Then a bird flies in and swirls around in the air to see if any danger is there.
- c. Then it sits on the apple picks it up and flies away.

2. Object Definition

1. A tree:



2. An apple:



3. Mud on the ground:



4. Bird:



Key frames:





General Computer Animation Functions:

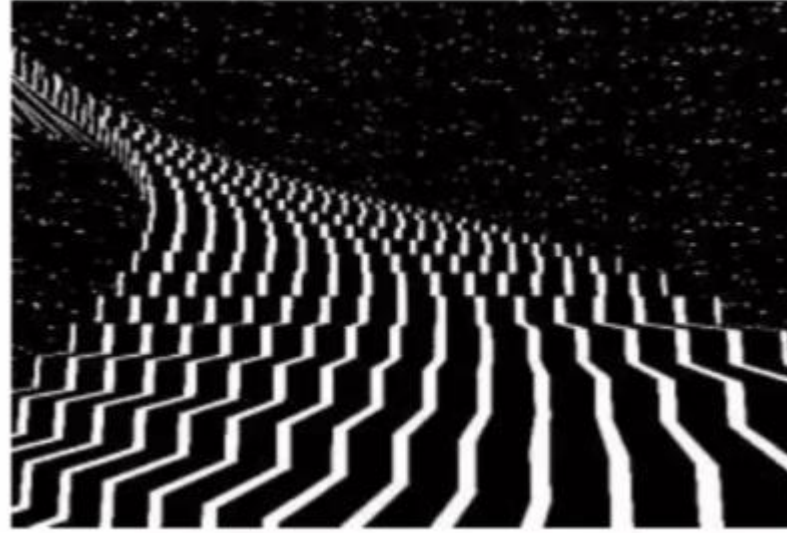
1. Store and manage the object database
 2. Functions for motion generation and for object rendering
 3. Functions that simulates camera movements
 4. Zooming, panning, and tilting
- Some steps in development of animation sequence are well - suited to computer solution. These include object manipulation and rendering camera motions, and the generation of in - betweens. One function available in animation packages is provided to store and manage the object database. Object shapes and associated parameters are stores and updated in the database. Other object functions include those for motion generation and those for object rendering. Motion can be generated according to specified constraints using two dimensional or three dimensional transformations. Another typical function simulates camera movements. Standard motions are zooming, panning, and tilting

Computer Animation Languages:

1. C
 2. Lisp
 3. Pascal
 4. FORTRAN
 5. Special animation packages like Adobe Flash, Autodesk, and maya etc.
- Design and control of animation sequences are handled with a set of animation outlines. A general purpose language, such as C, Lisp, Pascal or FORTRAN is often used to program the animation functions, but several specialized animation languages have been developed. Animation functions include a graphics editor, a key frame generator, an in-between generator and standard graphics routines. The graphics editor allows us to design and modify object shapes, using spline surface, constructive solid geometry methods, or other representation schemes.

Raster Animation

- ❑ On raster scan systems we can generate real time animation in limited applications using raster operations.
- ❑ A simple method for translation in the xy plane is to transfer a rectangular blocks of pixels through arbitrary angles using anti-aliasing procedures. To rotate a block of pixels we need to determine the percent of area coverage for those pixels that overlap the rotated block.
- ❑ Sequences of raster operations can be executed to produce real time animation of either two-dimensional or three-dimensional objects, as long as we restrict the animation to motions in the projection plane.
- ❑ Then no viewing or visible surface algorithms need be invoked. Color table transformations also produce animation. Just by changing the intensity values of pixels to on and off, we can produce animation on raster systems.



Real-time animations can be generated using raster operations.

Anti-Aliasing Methods:

A high-resolution display, post-filtering (super-sampling), pre-filtering (area sampling), and pixel phasing are the techniques used to remove aliasing. The explanations of these are given below:

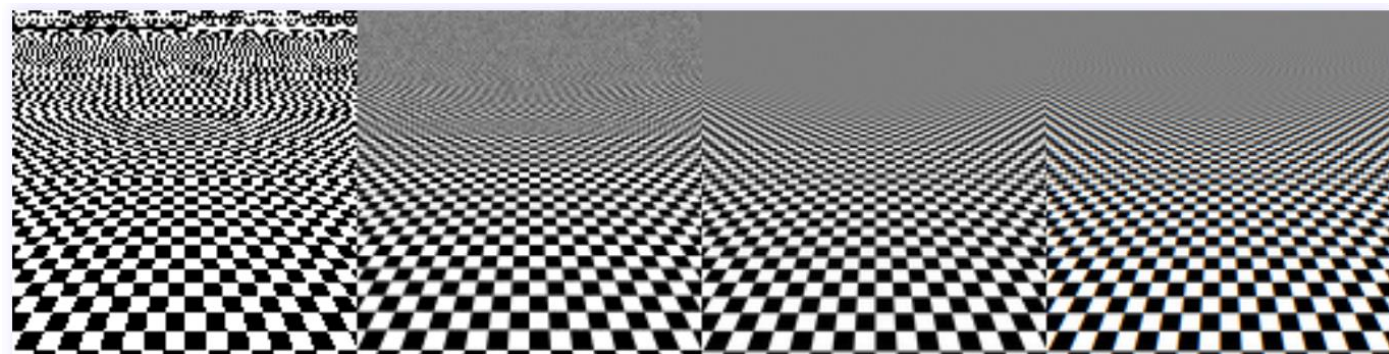
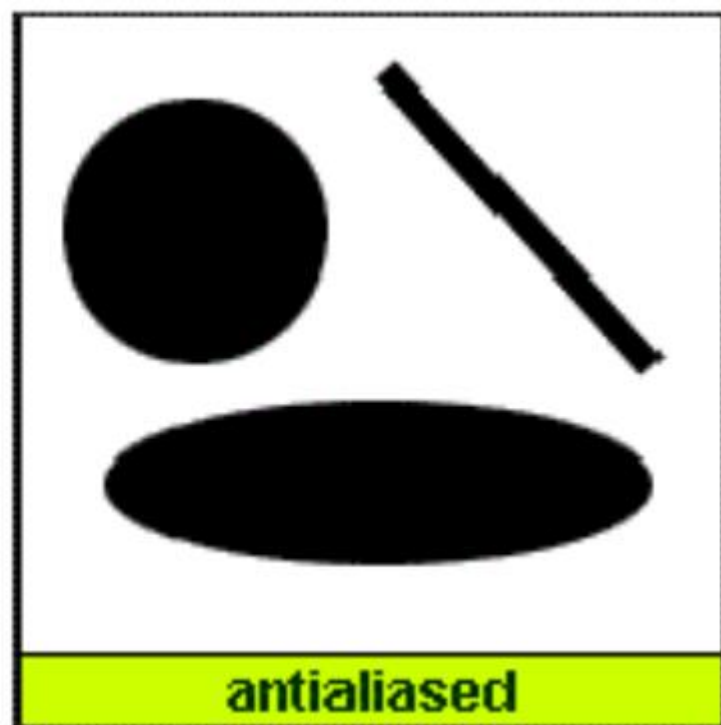
a. Using High-Resolution Display - Displaying objects at a greater resolution is one technique to decrease aliasing impact and boost the sampling rate. When using high resolution, the jaggies are reduced to a size that renders them invisible to the human eye. As a result, sharp edges get blurred and appear smooth.

- Real-Life Applications: For example, OLED displays and retina displays in Apple products both have high pixel densities, which results in jaggies that are so microscopic that they are blurry and invisible to the human eye.

b. Post-Filtering or Super-Sampling - With this technique, we reduce the adequate pixel size while improving the sampling resolution by treating the screen as though it were formed of a much finer grid. The screen resolution, however, does not change. Now, the average pixel intensity is determined from the average of the intensities of the subpixels after each subpixel's intensity has been calculated. In order to display the image at a lesser resolution or screen resolution, we do sampling at a higher resolution, a process known as super sampling. Due to the fact that this process is carried out after creating the rasterized image, this technique is also known as post filtration.

C. Pre-Filtering or Area-Sampling - The areas of each pixel's overlap with the objects displayed are taken into account while calculating pixel intensities in area sampling. In this case, the computation of pixel color is centered on the overlap of scene objects with a pixel region.

d. **Pixel Phasing** - It is a method to eliminate aliasing. In this case, pixel coordinates are altered to virtually exact positions close to object geometry. For dispersing intensities and aiding with pixel phasing, some systems let you change the size of individual pixels.



Aliased
image

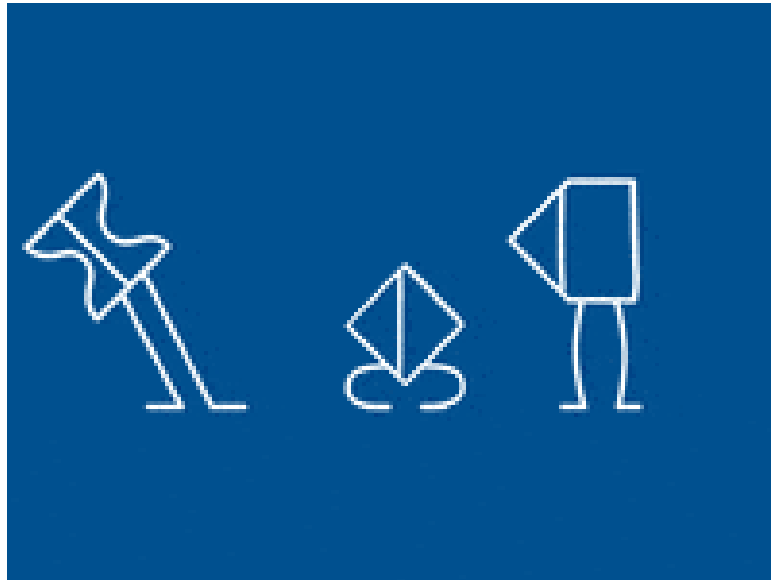
Pixel
Coverage
Anti-aliasing

Super
sampling
with proper
filtering

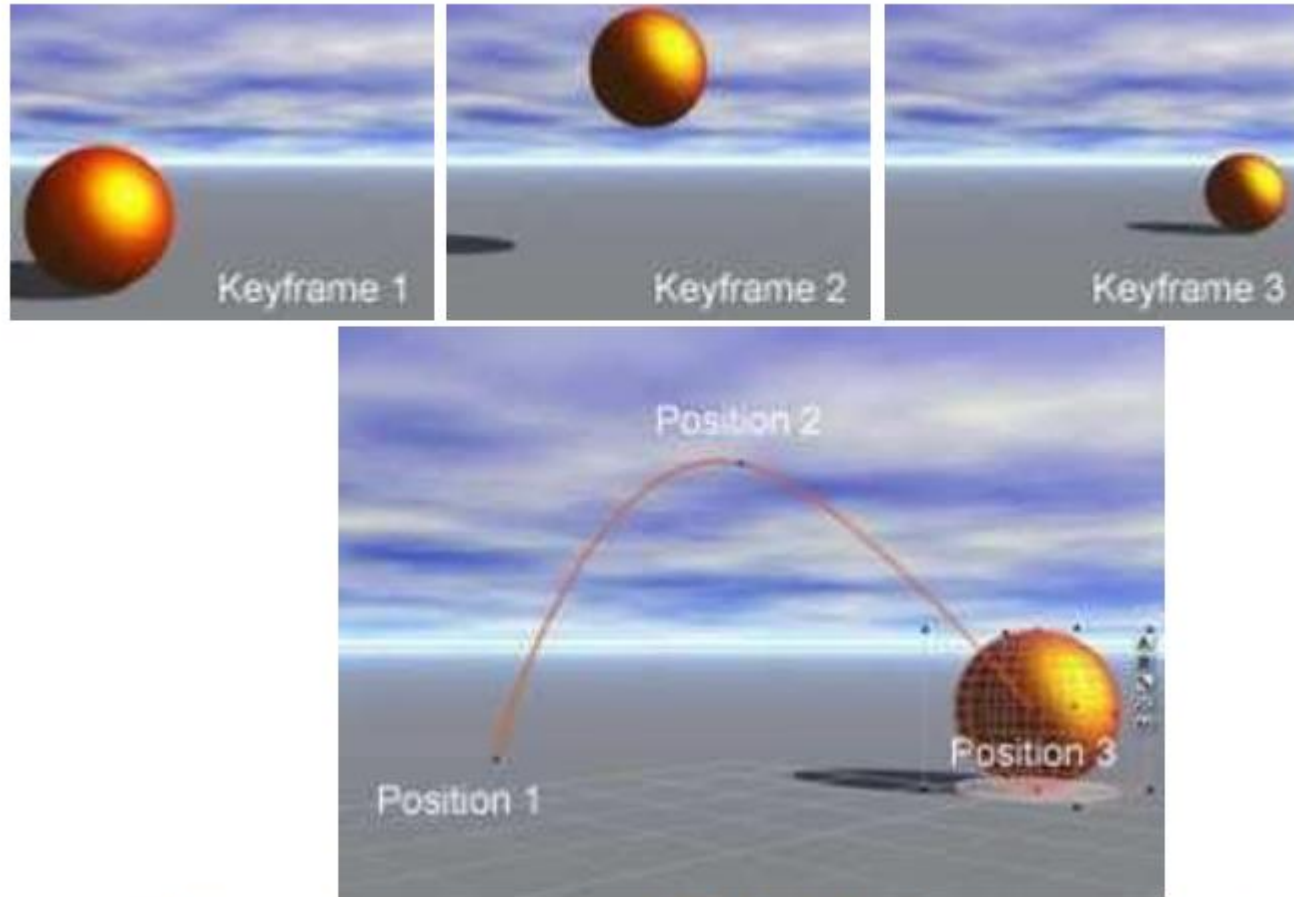
Same with
subpixel
rendering

keyframe system

- The **keyframe system** in animation is a technique where animators define the most important frames (keyframes) in a sequence of motion, and the in-between frames (called "in-betweens" or "tweens") are automatically generated or manually drawn to create a smooth transition from one keyframe to the next.
- Keyframes are the starting and ending points of any smooth transition or movement. Here's a breakdown of how the keyframe system works



- Keyframing is an animation technique where motion curves are interpolated through states at times, $(\sim q_1, \dots, \sim q_T)$, called keyframes, specified by a user



- **Key Components:**

- **Keyframes:**

- These are the most important frames that define the start and end points of a specific action or movement.
- They determine critical poses, expressions, or positions at specific moments in time.
- For example, in a walking animation, keyframes might include the position of the character's legs at the beginning of the step and at the end of the step.

- **In-Between Frames (Tweens):**

- These are the frames that occur between the keyframes.
- These frames are created to ensure smooth transitions and to "fill in the gaps" between keyframes.
- In traditional animation, animators would draw these frames manually. In digital animation, they can often be generated by the software (known as "tweening").

- **Timing and Spacing:**

- Timing refers to the number of frames used between key frames to define how fast or slow the animation appears.
- Spacing refers to how the position of an object changes between frames. If the spacing between frames is large, the movement appears fast. If the spacing is small, the movement appears slow and fluid.

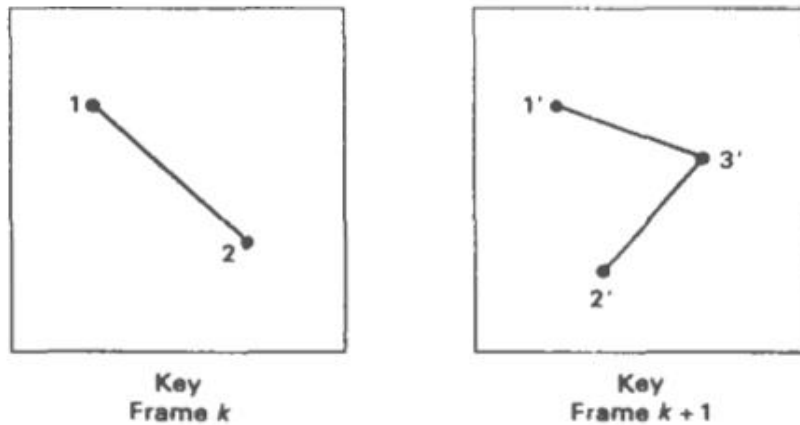
How It Works:

- **Planning:**
 - ❑ The animator starts by defining the major poses and key moments of the action. These are the keyframes. For example, in an animation of a bouncing ball, the keyframes could be the ball at the top of its bounce and at the lowest point of its drop.
- **In-Betweening (Tweening):**
 - ❑ After the key frames are established, the in-between frames are created to smooth the transition from one key frame to another.
 - ❑ These can be manually drawn (traditional animation) or generated by software (in digital animation, tools like **Adobe Animate** and **Blender** can interpolate the in-between frames based on the key frames).
- **Adjusting Timing and Spacing:**
 - ❑ The animator adjusts the timing and spacing between key frames to fine-tune how the movement feels. For example, a character's movement might slow down when they reach the apex of a jump and then speed up again as they fall.
- **Polishing:**
 - ❑ After the keyframes and in-betweens are set, the animator may refine the animation further by adding more detailed in-betweens, adjusting the timing, or making changes to the keyframes.

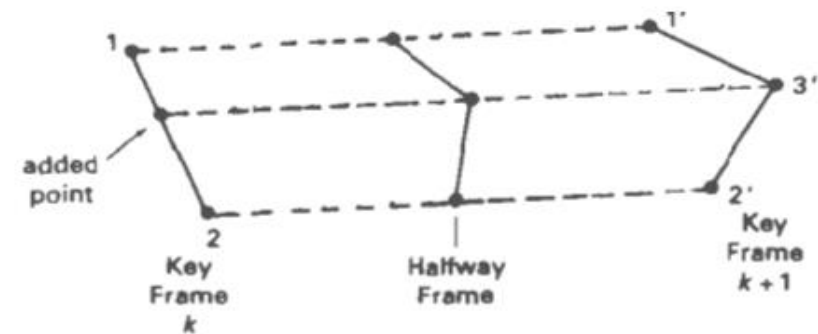
- **Pros:** – Very expressive
 - Animator has complete control over all motion parameters
- **Cons:** – Very labor intensive
 - Difficult to create convincing physical realism
- **Uses:** – Potentially everything except complex physical phenomena such as smoke, water, or fire

Morphing

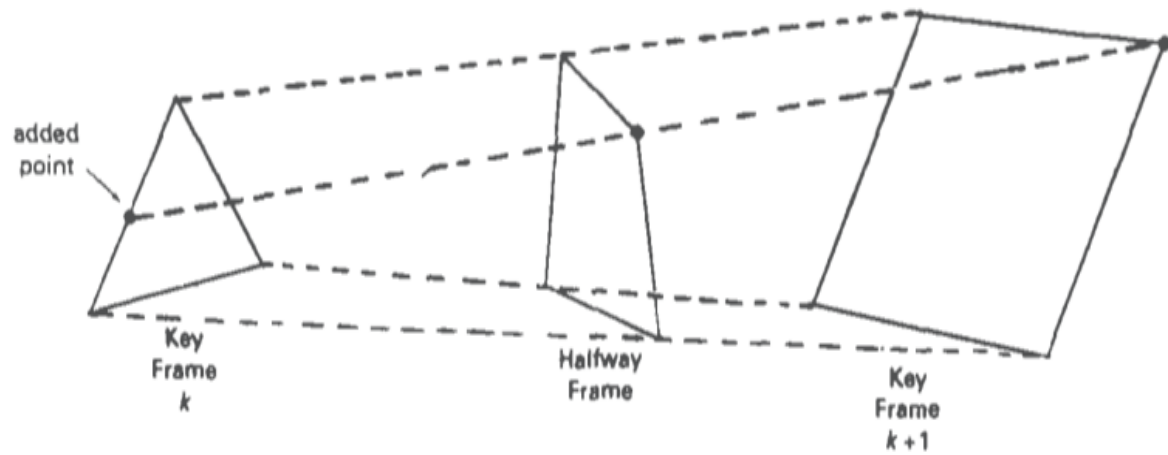
- ❑ Transformation of object shapes from one form to another is called morphing, which is a shortened form of metamorphosis.
- ❑ Morphing methods can be applied to any motion or transition involving a change in shape.
- ❑ Given two key frames for an object transformation, we first adjust the object specification in one of the frames so that the number of polygon edges (or the number of vertices) is the same for the two frames.
- ❑ This preprocessing step is illustrated in Figure.



An edge with vertex positions 1 and 2 in key frame k evolves into two connected edges in key frame $k + 1$.



Linear interpolation for transforming a line segment in key frame k into two connected line segments in key frame $k + 1$.



Linear interpolation for transforming a triangle into a quadrilateral.

For Example:

1. Human Face is converted into animal face as shown in fig:



2. Face of Young person is converted into aged person as shown in fig:



In computer graphics, **morphing** refers to a smooth transformation or blending of one image or shape into another over a series of frames. It's commonly used in animations, visual effects, and video games to create seamless transitions. Here's **an outline of the typical process of morphing**:

1. Selection of Source and Target Shapes or Images

- ❑ First, two objects or images are selected: the **source** (initial shape or image) and the **target** (final shape or image).
- ❑ These can be simple geometric shapes, complex objects, or even images.

2. Correspondence Mapping (Feature Correspondence)

- ❑ This step involves defining corresponding points, features, or control points on both the source and target images. These are usually key features like edges, corners, or other landmarks.
- ❑ For example, if morphing faces, the eyes, nose, and mouth would be the critical points to match up.

3. Warping

- **Warping** refers to the process of transforming the source shape or image into the shape of the target. This transformation is done by adjusting each point on the source to match the corresponding point on the target.
- Geometric transformations like scaling, rotating, and translating are used, often involving more complex interpolation techniques such as **affine transformation** or **non-linear warping**.

4. Interpolation

- Once the correspondence is defined, interpolation is used to blend the features or points between the source and the target.
- Common methods of interpolation are:
 - **Linear interpolation**: A straightforward method where each feature gradually moves from its source position to its target position.
 - **Spline-based interpolation**: A smoother method that can create a more natural-looking transition.
 - **Bezier curves** or **B-splines** can be used to define the interpolation paths for more complex or organic shapes.

5. Cross-Dissolving (Blending)

- After warping, **color and texture blending** happens to make the transition between the images smooth.
- Pixel values of the source image are gradually blended with the target image using alpha blending, where a range of opacity is applied to each frame. This creates a gradual fade from one image to the other.

6. Animation

- To create the morph, multiple intermediate frames are generated between the source and target images. These frames represent the gradual transition from the source to the target.
- Typically, the number of intermediate frames determines the smoothness of the morph. More frames mean a smoother transition.
- The morphing sequence is then played as an animation, where each frame smoothly transitions from one image/shape to the next.

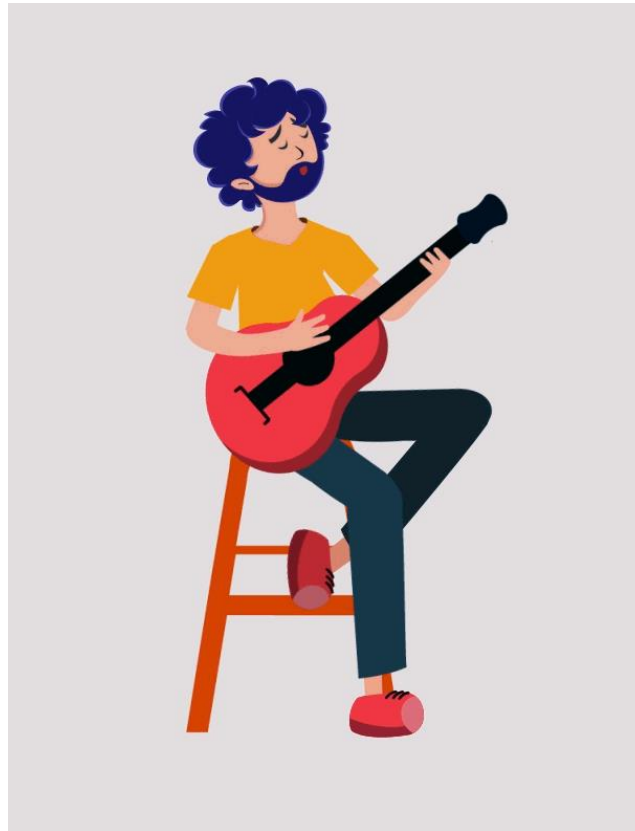
7. Refinement

- To enhance the morph, additional adjustments can be made:
 - Fine-tuning the feature correspondences for more accuracy.
 - Using advanced algorithms like **multi-resolution morphing** or **image pyramids** to improve the quality.
 - Sometimes, **texture mapping** can be applied to ensure that the texture of the source image smoothly blends with the target image.

MOTION SPECIFICATIONS

- ❑ There are several ways in which the motions of objects can be specified in an animation system.
- ❑ We can define motions in very explicit terms, or we can use more abstract or more general approaches.
- ❑ Motion graphics can be simple, such as animating text to appear on screen, or complex, such as creating 3D animations or using special effects.
- ❑ It can be used to create a wide variety of different looks and styles that can fit across needs of all industries and global enterprises. It can be used to communicate complex information in a way that is easy to understand for common people as it's easy to see than to hear.

- ❑ Motion graphics are the reason why science fiction films are possible to develop and that to on a very large scale.
- ❑ Without motion graphics animation industry wouldn't have progressed as far as they are today.
- ❑ A very famous example is GIF which we use in day-to-day texting.

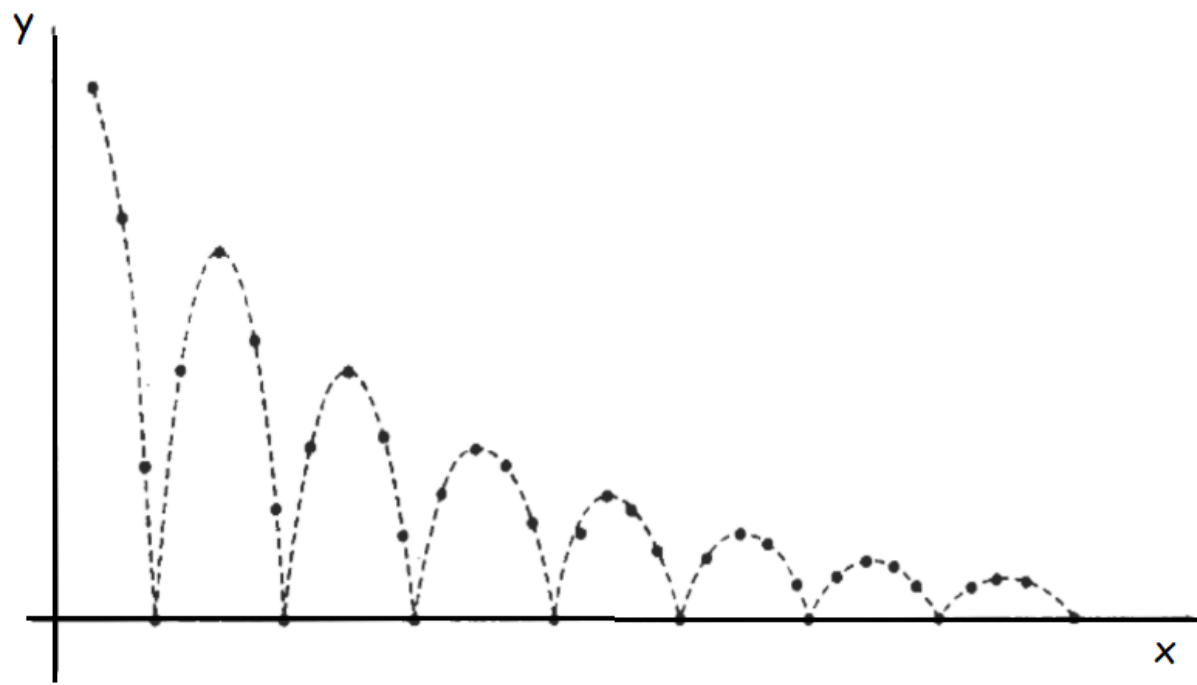


1. Direct Motion Specification

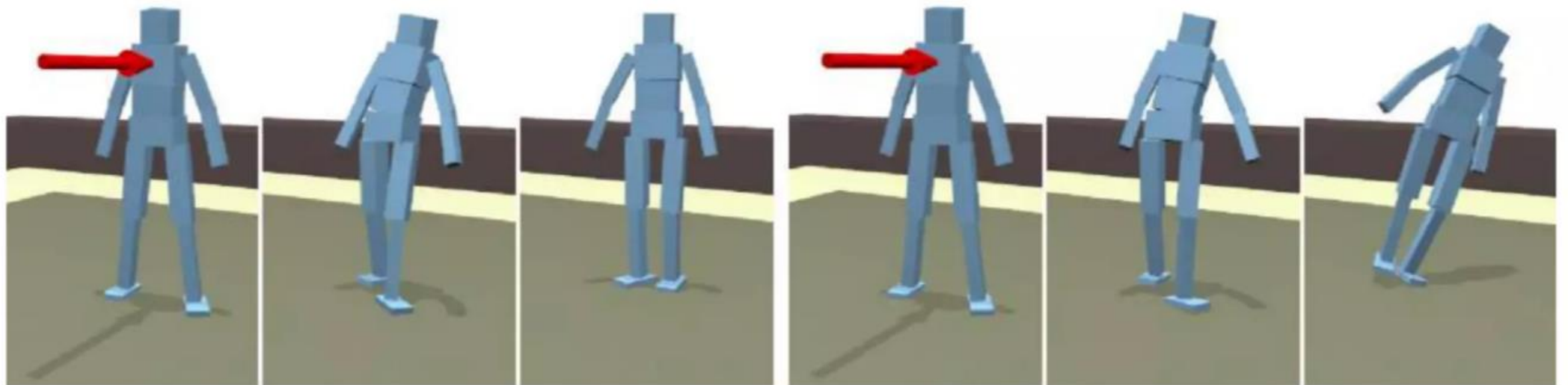
- ❑ The most straightforward method for defining a motion sequence is direct specification of the motion parameters.
- ❑ Here, we explicitly give the rotation angles and translation vectors.
- ❑ Then the geometric transformation matrices are applied to transform coordinate positions.
- ❑ Alternatively, we could use an approximating equation to specify certain kinds of motions.
- ❑ We can approximate the path of a bouncing ball, for instance, with a damped, rectified, sine curve (figure)

$$Y(x) = A | \sin (wx + \theta_0) | e^{kx}$$

- ❑ where A is the initial amplitude, w is the angular frequency, θ_0 , is the phase angle, and k is the damping constant



Goal Directed System



2. Goal-Directed Systems

- ❑ At the opposite extreme, we can specify the motions that are to take place in general terms that abstractly describe the actions.
- ❑ These systems are referred to as goal directed because they determine specific motion parameters given the goals of the animation.
- ❑ For example, we could specify that we want an object to "walk" or to "run" to a particular destination.
- ❑ Or we could state that we want an object to "pick up" some other specified object.
- ❑ The input directives are then interpreted in terms of component motions that will accomplish the selected task.
- ❑ Human motions, for instance, can be defined as a hierarchical structure of sub-motions for the torso, limbs, and so forth.

3. Kinematics and Dynamics

- ❑ We can also construct animation sequences using kinematic or dynamic descriptions.
- ❑ With a kinematic description, we specify the animation by giving motion parameters (position, velocity, and acceleration) without reference to the forces that cause the motion.
- ❑ For constant velocity (zero acceleration), we designate the motions of rigid bodies in a scene by giving an initial position and velocity vector for each object.
- ❑ As an example, if a velocity is specified as $(3, 0, -4)$ km/sec, then this vector gives the direction for the straight-line motion path and the speed (magnitude of velocity) is 5 km/sec.
- ❑ If we also specify accelerations (rate of change of velocity), we can generate speed-ups, slowdowns, and curved motion paths.
- ❑ Kinematic specification of a motion can also be given by simply describing the motion path.
- ❑ This is often done using spline curves.

- ❑ An alternate approach is to use [inverse kinematics](#).
- ❑ Here, we specify the initial and final positions of objects at specified times and the motion parameters are computed by the system.
- ❑ For example, assuming zero accelerations, we can determine the constant velocity that will accomplish the movement of an object from the initial position to the final position.
- ❑ This method is often used with complex objects by giving the positions and orientations of an end node of an object, such as a hand or a foot.
- ❑ The system then determines the motion parameters of other nodes to accomplish the desired motion

- **Dynamic descriptions** on the other hand, require the specification of the forces that produce the velocities and accelerations.
- Descriptions of object behavior under the influence of forces are generally referred to as a physically based modeling.
- Examples of forces affecting object motion include electromagnetic, gravitational, friction, and other mechanical forces. Object motions are obtained from the force equations describing physical laws, such as Newton's laws of motion for gravitational and friction processes, Euler or Navier-Stokes equations describing fluid flow, and Maxwell's equations for electromagnetic forces.
- For example, the general form of Newton's second law for a particle of mass m is

$$f = \frac{dv}{dt}(mv) \text{ With } F \text{ as the force vector, and } v \text{ as the velocity vector}$$

- ❑ If mass is constant, we solve the equation $F = ma$, where a is the acceleration vector.
- ❑ Otherwise, mass is a function of time, as in relativistic motions or the motions of space vehicles that consume measurable amounts of fuel per unit time.
- ❑ We can also use inverse dynamics to obtain the forces, given the initial and final positions of objects and the type of motion.

Thank you